

Project Elara

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Client View



Engaging RPG Experience

Immersive Gameplay: Players journey through a mysterious labyrinth to rescue Elara

Progressive Difficulty: 50 unique rooms with scaling challenges

Rich Equipment System: Over 20 unique items including weapons, armor, and potions

Strategic Depth: Currency management and equipment choices matter



Player Progression & Features

Dynamic Combat System: Tactical decisions with armor damage reduction and weapon upgrades

Multiple Game Systems:

- Engaging shop system with multiple tiers of equipment
- Mathematical challenges for added variety
- Random events that affect player health
- Epic boss battle with unique dodge mechanics

Balanced Economy: Starting with 15 coins, players make strategic purchase decisions



Player Engagement Metrics

Deep Item Variety:

- 5 tiers of weapons from Stick Sword to legendary Shadow Sword
- Multiple armor types with increasing protection values
- 5 potion types offering different healing amounts

Progression Milestones: New equipment unlocks at rooms 11, 21, and 41

Final Boss Challenge: Timer-based combat with pattern recognition



Demo

```
=====
Get re... No one said it would ever be easy....
I
=== Current Status ===
Room:
HP: 100
Coins: 15
Weapon: Basic Sword (Damage: 5)
Arm|
```



Technical Side



Core Architecture

- Player State Management - HP/Combat calculations, Inventory tracking, Equipment management

Shop System - Tiered item availability, Purchase validation, Duplicate prevention

Combat Engine - Damage calculation, Armor reduction, Random number generation



Testing Coverage

- **70+ Unit Tests** covering:
 - Player state initialization
 - Equipment management
 - Combat calculations
 - Shop functionality
 - Edge cases
- **Systematic Validation:**
 - Input boundary testing
 - State transition verification



Technical Features

Combat System:

- Dynamic damage calculation with random variation
- Armor reduction formula: $\text{damage} * (1 - \text{armorReduction})$
- Weapon damage scaling from 5 to 50

State Management:

- Persistent inventory tracking
- Purchase history validation
- Room-based progression system

Boss Fight Implementation:

- Deterministic pattern generation using seed
- Timer-based mechanics
- State machine for attack sequences



Final Boss Battle

BOSS ATTACK DIRECTION: RIGHT

Wrong directi

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